

Table 2: Comparison of N_{GE} for different transfer learning methods across homogeneous and heterogeneous settings.

Homogeneous (Cross Device) N_{GE}								
Source Device	Target Device	Direct Profiling (N_{GE}^{DP})	Direct Transfer (N_{GE}^{DT})	Unsupervised Transfer Learning Attacks (N_{GE}^{US})				
				CDPA* [CZLG21]	ALPA* [CZG+22]	Dual-Leak\$ [YWS+23]	On-the-fly FT\$ [RAID24]	UTLA (Ours)
XMEGA _{pow}	XMEGA _{pow}	12	357	15	17	16	15	16
XMEGA _{EM}	XMEGA _{EM}	14	737	58	57	55	65	17
SAKURA-AES	SAKURA-AES	586	1640	1425	1356	1524	1456	716
Heterogeneous (Cross Domain) N_{GE}								
SAKURA-AES	XMEGA _{pow}	12	N.C.	3210	4123	N.C.	N.C.	61
XMEGA _{pow}	SAKURA-AES	586	N.C.	3892	4533	N.C.	N.C.	788
SAKURA-AES	XMEGA _{EM}	14	N.C.	4474	4895	N.C.	N.C.	142
XMEGA _{EM}	SAKURA-AES	586	N.C.	4884	N.C.	N.C.	N.C.	841
XMEGA _{EM}	XMEGA _{pow}	12	N.C.	1870	2980	3285	3774	47
XMEGA _{pow}	XMEGA _{EM}	14	N.C.	2494	3480	3153	3485	56
ASCADv1 ^{&} (Random)	ASCADv1 ^{&} (Fixed)	250	N.C.	N.C.	N.C.	N.C.	N.C.	1330
ASCADv2 ^{&}	ASCADv1 ^{&} (Fixed)	250	N.C.	N.C.	N.C.	N.C.	N.C.	1519
XMEGA _{pow}	Cache Side Channel	187	N.C.	N.C.	N.C.	N.C.	N.C.	435
Cache Side Channel	XMEGA _{pow}	195	N.C.	N.C.	N.C.	N.C.	N.C.	1233
SAKURA-AES	Cache Side Channel	187	N.C.	N.C.	N.C.	N.C.	N.C.	1424
Cache Side Channel [†]	SAKURA-AES [†]	195	N.C.	N.C.	N.C.	N.C.	N.C.	N.C.

Remarks – N.C.: Training did not converge to $GE = 1$, which means that the attack cannot find the target key.; *: Attacks adapted from the codebase provided in the paper; \$: In-house adaptation of the active learning algorithm. Highlighted values corresponds to the new results beyond the initial submission. &: For ASCAD, only ASCADv1(Random) to ASCADv1(Fixed) and ASCADv2 to ASCADv1(Fixed) transfers lead to converging results, as in all other cases, i.e., transfer to and from XMEGA/SAKURA/Cache to ASCAD have misaligned encoder outputs. Likewise, Cache Side Channel[†] to SAKURA-AES[†] did not converge in the case of UTLA. Note that UTLA succeeds when the source domain shares transferable side-channel information with the target, corresponding to aligned features at the encoder outputs.